UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	PRE CALCULUS
COURSE CODE	MAT133
EXAMINATION	OCTOBER 2026
TIME	2 HOURS

NSTRUCTIONS TO CANDIDATES

1. This question paper consists of 5 questions.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) a one – page Appendix 1
   iii) an Answer Booklet - provided by the Faculty
   iv) a graph paper - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 9 printed pages

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QUESTION 1

a) Solve the compound inequality $-3 < 2x + 6 < 0$. Express the answer in interval notation.

(4 marks)



Figure: solution on a number line.

b) Solve the inequality $|5x + 3| \geq 4$. Express the answer in interval notation and on a number line.

(5 marks)



Figure: solution on a number line.

c) Solve the equation $|2x + 1| = 6$.

(4 marks)

QUESTION 2

a) Given $f(x) = (x - 3)/(x - 1)$ and $g(x) = \sqrt{(4x + 6)}$, state the domain of each function.

(4 marks)

b) Given $f(x) = (2 - 5x)/(3x - 4)$, $x \neq 4/3$, find $f^{-1}(x)$ and state its domain.

(5 marks)

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c) Sketch the circle $x^2 - 4x + y^2 + 2y - 11 = 0$. State its centre and radius.

(4 marks)

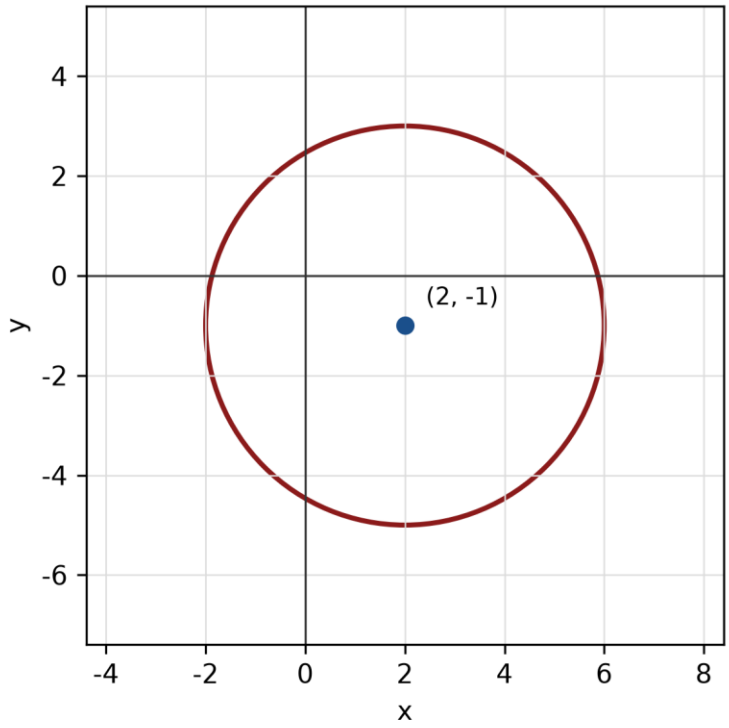


Figure: circle.

QUESTION 3

a) Given $z_1 = -5 - 2i$ and $z_2 = 4 - i$, find $z_1 + z_2$ in the form $a + bi$.

(4 marks)

b) Evaluate i^6 and simplify completely.

(3 marks)

c) Express log_5(5) + log_5(5) as a single logarithm and evaluate it.

(3 marks)

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QUESTION 4

a) Graph the system of inequalities below and shade the feasible region. State the vertices of the region.
$x \geq 0, y \geq 0, 2x + 1y \leq 8.$

(5 marks)

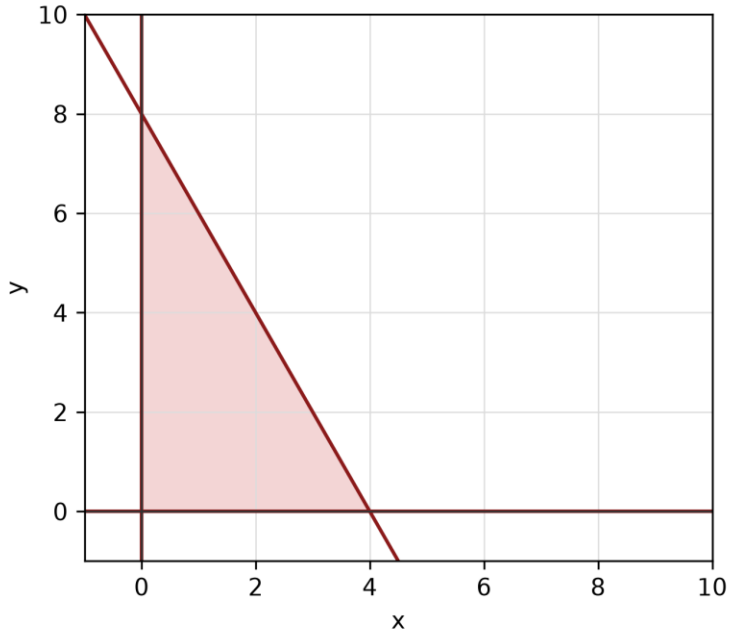


Figure: feasible region.

b) Solve the simultaneous equations
$y = x^2 - 3$
$y = 1.$

(5 marks)

c) Find the equation of the line passing through (2, 1) and parallel to the line $3x - 2y - 4 = 0$.

(5 marks)

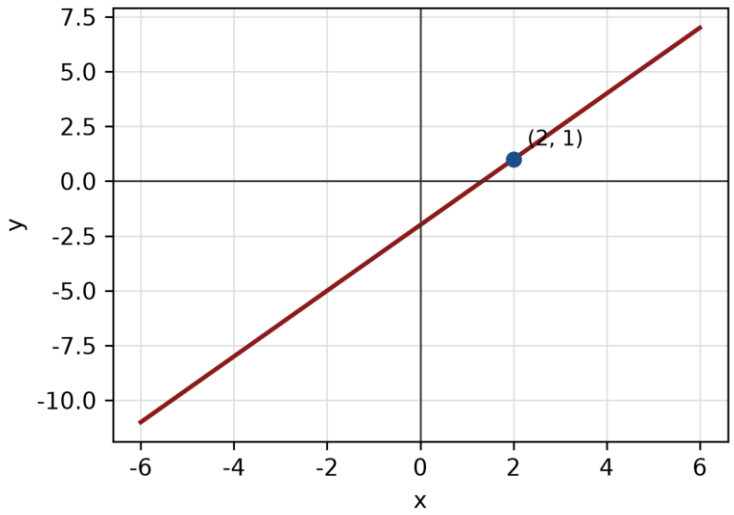


Figure: required line.

QUESTION 5

a) Prove the identity $(1 - \cos^2 x)/(\sin x \cos x) = \tan x$.

(5 marks)

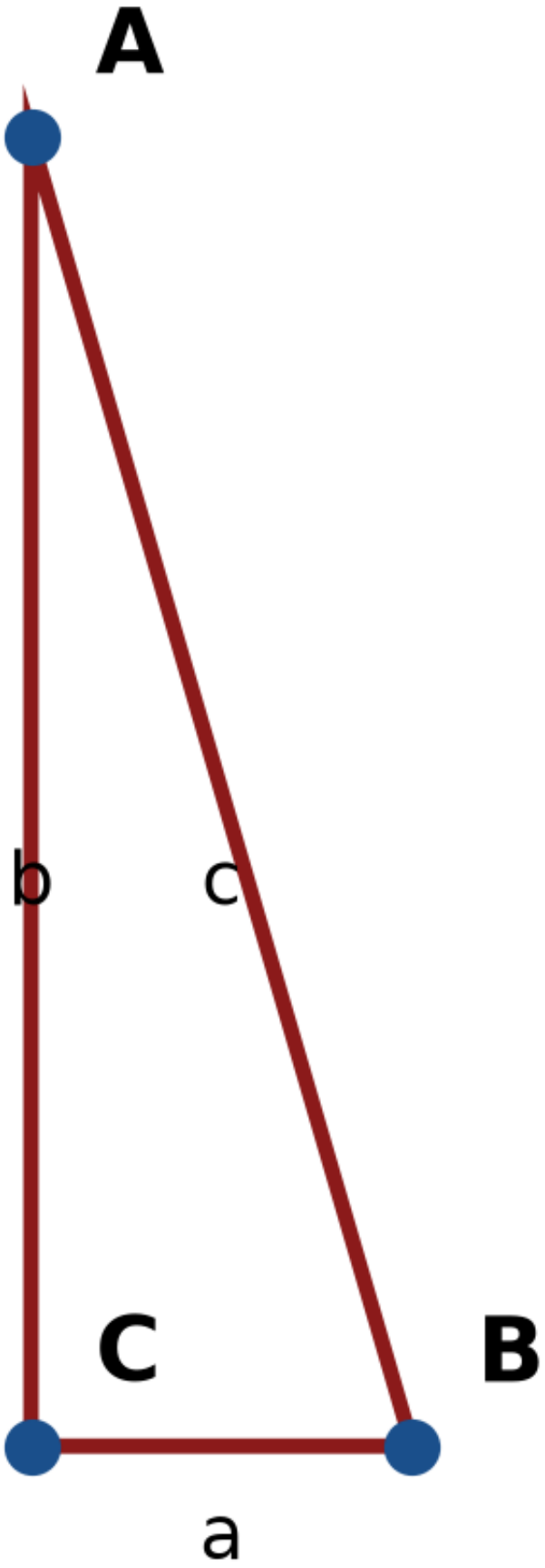
b) Convert 180° to radians, and convert π/4 radians to degrees.

(3 marks)

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c) Solve the triangle with sides a = 14 cm, b = 48 cm and c = 50 cm. Find all three angles.
(5 marks)

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Figure: right triangle.

END OF QUESTION PAPER

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APPENDIX 1

Formula sheet (as supplied with the paper).

$\sin^2 A + \cos^2 A = 1$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\tan(A - B) = (\tan A - \tan B) / (1 + \tan A \tan B)$

$\cos 2A = \cos^2 A - \sin^2 A$

$\cos 2A = 1 - 2\sin^2 A = 2\cos^2 A - 1$

$\sin 2A = 2 \sin A \cos A$

$\tan^2 A + 1 = \sec^2 A$

$\tan 2A = 2 \tan A / (1 - \tan^2 A)$

Arc length, $s = r\theta$

Area of sector = $\frac{1}{2}r^2\theta$

$a / \sin A = b / \sin B = c / \sin C$

$a^2 = b^2 + c^2 - 2bc \cos A$

Area of triangle = $\frac{1}{2}ab \sin C$

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